

1 Irrational numbers

Surds

2012-5.

Without using a calculator or four-figure tables, simplify $\frac{8\sqrt{2}}{\sqrt{98}-3\sqrt{2}}$, giving your answer in its simplest form.
(4 marks)

2011-3.

Without using a calculator or four-figure tables, simplify $\frac{1}{\sqrt{2}} - \frac{\sqrt{2}}{3}$ leaving your answer with a rational denominator. (4 marks)

2010-5.

Without using a calculator or four figure tables, simplify $\frac{\sqrt{54}+3\sqrt{3}}{\sqrt{3}}$ in its simplest form. (4 marks)

2009-14.

Without using a calculator or four figure tables, simplify $\frac{11}{5-\sqrt{3}}$, leaving your answer with a rational denominator. (4 marks)

2008-2.

Without using a calculator or four-figure tables, simplify $\sqrt{27} \times \sqrt{32}$, leaving your answer in surd form.
(3 marks)

2007-1.

Express $\frac{\sqrt{2}}{\sqrt{3}}$ with a rational denominator. (3 marks)

2006-23.

Simplify $\frac{x^2-2}{x-\sqrt{2}}$. (3 marks)

2005-2.

Without using a calculator or four-figure tables, simplify $\sqrt{125} + \sqrt{5} - \sqrt{45}$, leaving your answer in surd form.
(3 marks)

2004-14.

Simplify $\frac{\sqrt{2}}{\sqrt{2}-1}$. (4 marks)

2003-3.

Express $\frac{3}{\sqrt{2}}$ as a fraction with a rational denominator. (3 marks)

2 Exponential and logarithmic functions

Exponential equations

2009-7.

Solve the equation $3^y = \frac{9^y}{81}$. (5 marks)

Logarithmic equations

2011-4.

Given that $\log_x 6\frac{1}{4} = 2$, solve for x . (4 marks)

2009-11.

Given that $\frac{1}{2}\log_3 x = \log_3(6 - x)^{\frac{1}{2}}$, find the value of x . (5 marks)

2008-6.

Given that $\log_x(1/2) + \log_x 16 = 3$, find the value of x . (4 marks)

2004-19.

Solve the equation $\log_{10}(2m + 6) = \log_{10}(m - 1)$. (5 marks)

Evaluation of logarithms

2007-11.

Given that $\log_a 2 = 0.668$ and $\log_a 3 = 0.884$, evaluate $\log_a 12$. (5 marks)

2003-8.

Given $\log_a 2 = 0.6110$ and $\log_a 3 = 0.7039$, calculate $\log_a 6$. (4 marks)

Rules of logarithms

2012-11.

Given that $\log p - \log q = 2 \log r$, find p in terms of q and r . (5 marks)

2010-8.

Given that $\log_5 x + \log_5 y = 3 \log_5 q$. Show that $x = \frac{q^3}{y}$. (3 marks)

2006-20.

Given that $\log_{10} n - \log_{10} m = 2 \log_{10} h$, show that $n = mh^2$. (4 marks)

2005-8.

Given that $\log_m 27 = 3$, find m . (3 marks)

3 Statistics

Histogram

2004-20.

In **Figure 5**, the histogram is representing the ages of the people who attended a meeting.

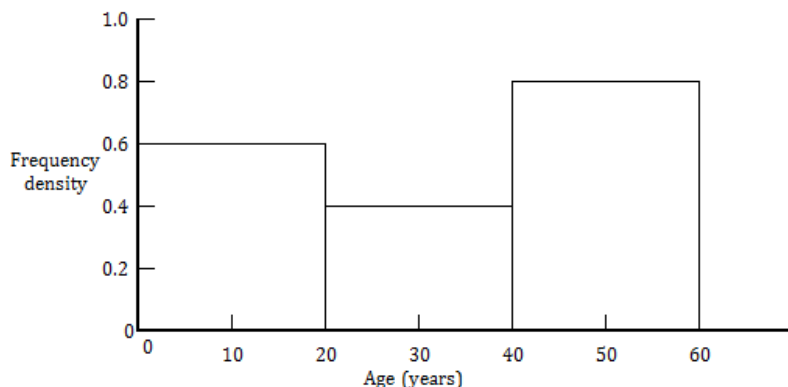


Figure 5

Calculate the number of people who were between 20 and 40 years old at the meeting. (3 marks)

2003-15.

Table 2 shows the distribution of the number of employees in 43 factories in a town.

Table 2

Number of Employees	0-39	40-59	60-79	80-99
Number of Factories	5	15	13	10

Draw a histogram. (4 marks)

Frequency polygon

2010-11.

The results of a test marked out of 25 written by 20 learners were as follows:

1	7	13	12	14
12	18	17	19	17
17	19	22	23	24
22	22	24	23	22

Using class intervals of 1 – 5, 6 – 10, 11 – 15, ..., construct a frequency table for the results. (3 marks)

2009-19.

Table 2 shows midpoints of marks scored by a group of 50 pupils in a test.

Table2

Midpoint mark	12	13	14	15	16	17	18	19
Number of pupils	2	3	6	12	10	8	6	3

Using a scale of 2 cm to represent 1 unit on the horizontal axis and 2 cm to represent 2 units on the vertical axis, draw a frequency polygon to represent this information. (5 marks)

Mean of data

2011-7.

The mean of $(x - 1)$, $(x + 2)$ and $(x + 5)$ is $2x$. Find the value of x . (5 marks)

2007-22.

Three data values x , y and z have the following relationship:

$$x = a^2 - a$$

$$y = 2 - a$$

$$z = 7 + 5a - a^2$$

Calculate the mean of x , y and z in terms of a in its simplest form. (3 marks)

2006-7.

Table 1 shows marks that student **A** and student **B** got from tests. Student **A** sat for 5 tests while student **B** sat for 4 tests. Student **B** has mark x missing.

Table 1

STUDENT A	55	70	80	30	65
STUDENT B	67	60	x	53	

Given that the mean mark of student **A** is the same as the mean mark of student **B**, calculate the value of x . (5 marks)

Variance

2005-16.

The table below shows ages of 5 pupils with the mean age of 12.6 years.

Age (yrs)	Deviation from mean	Square of deviation
10	-2.6	6.76
11	-1.6	2.56
13	0.4	0.16
14		
15	2.4	5.76
Total	0	

Copy and complete the table to calculate the variance of the ages. (5 marks)

Pie chart

2008-24.

In a survey conducted at Chitsa village, 20 people responded "YES", 30 responded "NO" and 10 responded "DON'T KNOW". Draw a pie chart to represent the information. (4 marks)

4 Matrices

Matrices

2012-10.

Given that $M = \begin{pmatrix} 3 & 0 \\ 1 & -2 \end{pmatrix}$, $N = \begin{pmatrix} -1 & 2 \\ 0 & k \end{pmatrix}$ and $= \begin{pmatrix} -1 & 2 \\ 0 & k \end{pmatrix}$, find the value of k . (5 marks)

2011-12.

T and R are two matrices. Given that $T = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$ and $R = \begin{pmatrix} 0 & 3 \\ -1 & 1 \end{pmatrix}$, find $3R - T^2$. (5 marks)

2010-7.

Given that matrix $P = \begin{pmatrix} 7 & 5 \\ 2 & 4 \end{pmatrix}$ and $Q = \begin{pmatrix} 3 & 10 \\ 5 & 1 \end{pmatrix}$, find PQ . (3 marks)

2009-23.

A and B are two matrices. If $A = \begin{pmatrix} 0 & 1 \\ 3 & 2 \end{pmatrix}$, find B given that $A^2 = A + B$. (3 marks)

2008-11.

Given that $\begin{pmatrix} 6 & 8 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} w & -4 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, find the value of w . (4 marks)

2007-9.

Given that $A = \begin{pmatrix} 2 & 6 \\ 1 & -4 \end{pmatrix}$, $B = \begin{pmatrix} -3 & 2 \\ 0 & -5 \end{pmatrix}$ and $C = \begin{pmatrix} -1 & 8 \\ -5 & 7 \end{pmatrix}$, simplify $\frac{1}{4}(A - B + C)$. (6 marks)

2006-11.

Find the values of x and y in the following matrix equation:

$$\begin{pmatrix} 6 & 3 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 12 \\ -4 \end{pmatrix}. \quad (5\text{marks})$$

2004-8.

Given that $\begin{pmatrix} 3c & c \\ 5 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 28 \\ 22 \end{pmatrix}$, find c . (4 marks)

2003-4.

Given that $a = \begin{pmatrix} 3 & 0 \\ -4 & 4 \end{pmatrix}$ and $b = \begin{pmatrix} 2 & -1 \\ -1 & 0 \end{pmatrix}$, calculate ab . (3 marks)

2003-6.

A point T has the coordinates $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$. The matrix which transforms T into T' is $\begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}$. Calculate the coordinates of T' . (3 marks)

5 Progressions

Arithmetic progression

2011-16.

The 3rd and 9th terms of an Arithmetic Progression are 29 and 8 respectively. Calculate the 20th term of the progression. (7 marks)

2009-12.

The 9th term of an arithmetic progression $y, y + 4, y + 8, \dots$ is 37. Find the value of y . (5 marks)

2008-8.

Find the sum of the first 20 terms of the arithmetic progression 4, 2, 0, ... (5 marks)

2007-14.

The 4th term of an arithmetic progression is 11 and the 7th term is 20. Calculate the first term. (5 marks)

2005-18.

The n^{th} term of an Arithmetic Progression is $5n - 3$. Calculate the sum of the first 6 terms of the AP. (5 marks)

Geometric progression

2012-17.

If the 4th and 7th term of a geometric progression (GP) are 4 and 32 respectively, calculate the **first** term. (7 marks)

2010-6.

A geometric progression has 6 terms. If its first term is 3 and the last term is 96, calculate the common ratio. (4 marks)

2006-22.

Find the sum of the first 12 terms of the following GP:

$$\frac{1}{2187}, \frac{1}{729}, \frac{1}{243}, \dots$$

Give your answer correct to 2 decimal places. (5 marks)

2004-9.

The sum of the first two terms of a geometric progression is 4. If the first term is 3, find the common ratio. (4 marks)

2003-17.

The first three terms of a G.P are $x + 1, x^2 - 1$ and $(x^2 - 1)(2x - 4)$. Calculate the value of x . (5 marks)

6 Probability

Probability

2012-16.

A box has 5 cards marked 1, 2, 3, 4 and 5. Another box has 2 cards marked A and B. If a card is picked randomly from each box at the same time, calculate the probability of picking a card marked 3 from one box and a card marked B from the other box. (4 marks)

2010-17.

Figure 6 is a tree diagram which shows the probability of picking two balls one at a time without replacement from a bag containing 6 red and 4 white balls.

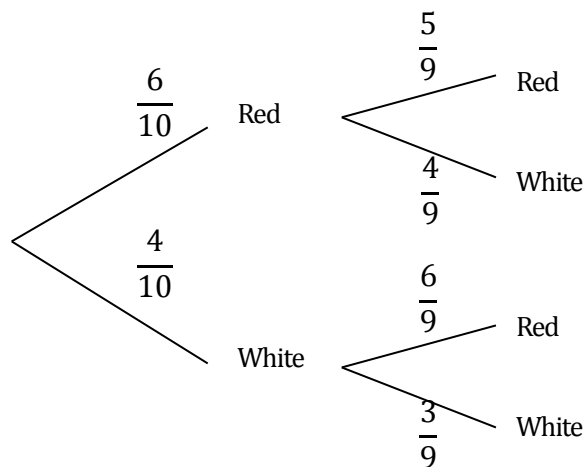


Figure 6

Use the tree diagram to calculate the probability of picking two balls of different colours, leaving your answer in its simplest form. (5 marks)

2009-9.

Table 1 shows the distribution of ages of learners in a Form 2 class.

Table 1

Age	14	15	16	17	18	19
Number of learners	2	10	8	4	9	3

What is the probability of picking at random a learner of 18 years of age? (3 marks)

2008-19.

Table 2 shows numbers. If a number is picked at random from the table, calculate the probability that it is prime or less than 9. (4 marks)

Table 2

16	17	3	13
5	11	10	8
9	7	6	12
4	14	15	17

2007-24.

The probability of having an early lunch at a boarding school is $\frac{2}{3}$. When the lunch is early, the probability of having beef is $\frac{7}{10}$ and when late, the probability of having beef is $\frac{1}{8}$. Draw a tree diagram to represent this information completing all branches. (4 marks)

2006-21.

Coin **A** is tossed followed by coin **B**. The probability that coin **A** shows head is $\frac{1}{2}$ while the probability that coin **B** shows head is $\frac{1}{4}$. Using a tree diagram, calculate the probability that both coins **A** and **B** show tails. (4 marks)

2005-21.

Figure 5 is a diagram illustrating the probability of a student passing Agriculture and Geography in an examination. The probability of passing Geography in the examination is $\frac{7}{10}$ and the probability of passing Agriculture after one has passed Geography is $\frac{5}{7}$. The probability of passing Agriculture after one failed Geography is $\frac{1}{6}$.

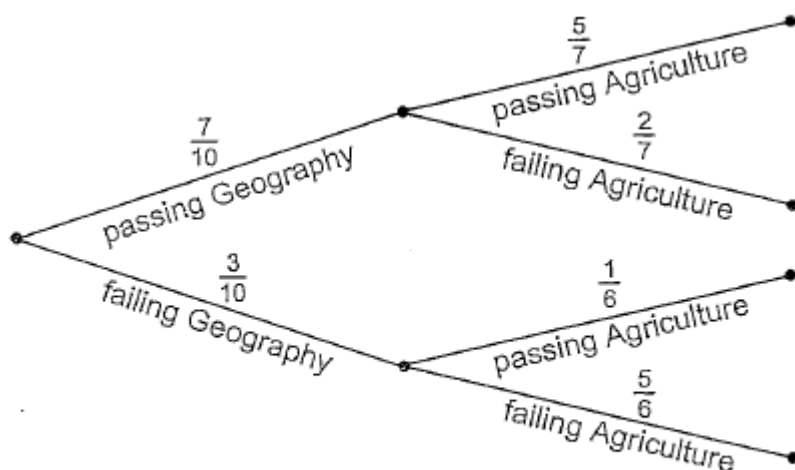


Figure 5

Calculate the probability of a student passing Agriculture. (5 marks)

2004-23.

In a plastic bag there are x blue pens, 6 black pens and 4 red pens. If the probability of picking a red pen is $\frac{1}{5}$, calculate the number of blue pens. (4 marks)

2003-14.

A box contains 5 red balls, 8 white balls and 7 black balls. If one ball is selected at random, calculate the probability that it is white or black. (5 marks)

7 Sets

Union and intersection of two or three sets

2012-13.

Given that P and Q are two sets such that $n(P \cup Q) = 37$, $n(P) = 30$ and $n(P \cap Q) = 5$, find $n(Q)$.
(4 marks)

2010-2.

Given that $X = \{a, c, e\}$, $Y = \{b, c, d, e\}$ and $Z = \{c, d, e, f\}$, find $(X \cup Y) \cap Z$. (3 marks)

2009-6.

In a family, six members eat meat, five members eat fish while two members eat both. Calculate the number of members in the family. (4 marks)

2008-18.

Given that $n(x) = 18$, $n(y) = 24$ and $n(x \cup y) = 40$. Find $n(x \cap y)$. (3 marks)

2005-5.

Given that $\{a, c, e, h, i, l, m, s, t, w\}$ is a universal set and $Y = \{a, c, e, h, i, m, s\}$, find $n(Y')$. (3 marks)

2004-22.

Given that the Universal set $\xi = \{11, 14, 15, 17, 18, 20, 23, 26\}$, set $X = \{11, 14, 15, 17, 18, 20\}$ and set $Y = \{15, 17, 18, 20, 23, 26\}$, find $X' \cup Y'$. (4 marks)

2003-5.

The universal set $(\xi) = \{10, 20, 30, 40, 50, 60, 70\}$, $A = \{10, 30, 60\}$ and $B = \{20, 40, 50\}$, evaluate $A' \cap B$. (3 marks)

Venn diagrams

2011-11.

Figure 2 shows a Venn diagram.

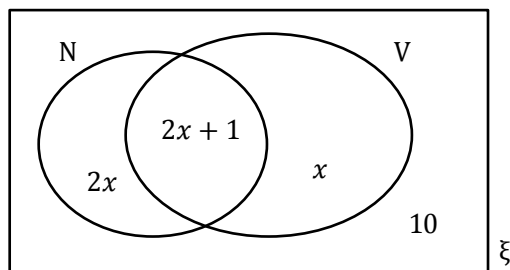


Figure 2

In the Venn diagram,

$\xi = \{\text{girls in form three}\}$

$N = \{\text{girls that play netball}\}$

$V = \{\text{girls that play volleyball}\}$

Given that there are 21 girls in the class, find how many girls play both netball and volleyball. (5 marks)

2007-8.

Figure 2 is a venn diagram showing the number of elements in sets **R**, **S** and universal set ξ .

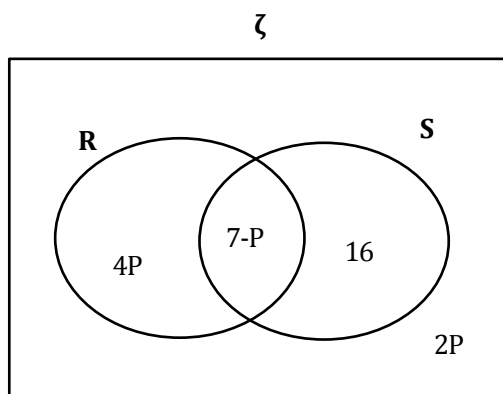


Figure 2

If $n(R \cup S) = 29$, calculate the value of p . (4 marks)

2006-15.

Figure 2 shows a venn diagram representing set of all numbers (**N**), set of even numbers (**E**), set of odd numbers (**O**) and set of prime numbers (**P**).

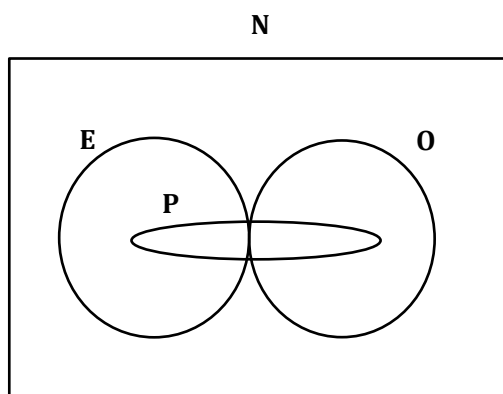


Figure 2

Copy the venn diagram and place the numbers $\frac{1}{2}$, 2, 6, 9 and 13 in the right places. (6 marks)

8 Quadratic expressions and equations

Factorization

2012-2.

Factorise $2m^2n - 3mn - 5n$ completely. (4 marks)

2011-1.

Factorise completely $3 - 5x - 2x^2$. (3 marks)

2010-1.

Factorise completely $2x^2 + 3xy - 35y^2$. (3 marks)

2009-1.

Factorise completely $2x^2 - 15x + 28$. (3 marks)

2008-1.

Factorise completely $3x^2y + 5xy + 2y$. (3 marks)

2007-3.

Factorise completely $2x^2 - 4x - 126$. (3 marks)

2006-2.

Factorise completely $2x^2 + 4xy - 30y^2$. (3 marks)

2005-1.

Factorise completely $3t^2 - 4t + 1$. (3 marks)

2004-1.

Factorise completely $6 + x - 2x^2$. (4 marks)

2003-1.

Factorise completely $x^2 + 3x + 4(x + 3)$. (3 marks)

Quadratic equations

2008-10.

Solve the equation $3(a + 1)^2 - 3 = 0$. (5 marks)

2006-8.

John is twice as old as Mary. If the sum of the squares of their ages is 125, how old is Mary? (6 marks)

2003-21.

Figure 2 shows a rectangular box with an open top. The box measures 6cm long, $2x$ cm wide and x cm high.

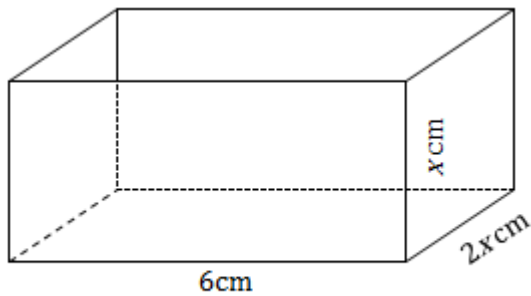


Figure 2

Given that the total outer surface area of the box is 108cm^2 , Form an equation in x and show that it simplifies to $x^2 + 6x - 27 = 0$. (5 marks)

Simultaneous linear and quadratic equations

2008-14.

Solve the simultaneous equations: (5 marks)

$$y = x - 2$$

$$xy + 1 = 0$$

2007-15.

Solve the simultaneous equations: (5 marks)

$$y = x + 2$$

$$x^2 - xy = 4$$

2005-24.

Solve the simultaneous equations (5 marks)

$$xy = -9$$

$$y = x + 6$$

2004-24.

Solve the simultaneous equations (5 marks)

$$y = x^2$$

$$y = 5x - 6$$

9 Algebraic fractions

Fractions with linear or quadratic denominators

2012-3.

Simplify $\frac{3}{y} - \frac{y-1}{y(y+2)}$. (5 marks)

2011-8.

Simplify $\frac{m}{m+2} - \frac{6}{m^2+m-2}$. (5 marks)

2010-15.

Simplify $\frac{d-1}{3} - \frac{2d+1}{7}$. (4 marks)

2009-16.

Express $\frac{1}{x-1} + \frac{3y}{xy-y}$ as a single fraction. (3 marks)

2008-3.

Simplify $\frac{x^2-1}{x} \times \frac{x^2}{x-1}$. (3 marks)

2007-10.

Simplify $\frac{1}{a-b} + \frac{1}{a+b}$. (4 marks)

2006-1.

Simplify the following fraction: $\frac{3x+6}{(x-1)(x+2)}$. (3 marks)

2005-4.

Simplify $\frac{x^2-y^2}{x^2-xy}$. (3 marks)

2004-16.

Express $\frac{m-2}{m-3} + \frac{m+3}{m+2}$ as a single fraction in its lowest term. (5 marks)

2003-10.

Express $\frac{1}{x^2-x-2} - \frac{1}{x+1}$ as a single fraction. (4 marks)

10 Mapping and functions

Domain and range

2012-1.

Given that $f(x) = bx + 4$ and $f(4) = 24$, find b . (5 marks)

2011-2.

Given that $g(x) = \frac{2x^2}{3} + 1$, find $g(-1)$ in its simplified form. (3 marks)

2010-4.

The function $f(y) = 3y + 2$. Given that $\{5\}$ is the range, find the domain. (3 marks)

2009-2.

Given that $f(x) = \frac{3}{3-x}$, find $f(-3)$ in its simplest form. (3 marks)

2007-4.

The function $f(x) = \frac{1}{3x-1}$. Given that $\{-1, 0, 2\}$ is the domain, find the range. (4 marks)

2006-4.

If $f(x) = 8^x - 6$, find $f(\frac{2}{3})$. (4 marks)

2005-6.

The function $y = 2 + x$ has the range $\{3, 6\}$. Find its domain. (3 marks)

2004-3.

Given that $g(x) = \frac{3x}{x+1}$, calculate the value of x when $g(x) = 2$. (4 marks)

2003-2.

Given that $f(x) = x^3 - x$, calculate $f(-2)$. (3 marks)

Mapping and arrow diagrams

2008-7.

Figure 2 shows an arrow diagram for the function $f(x) = 2x - 5$.

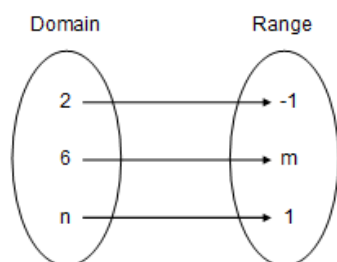


Figure 2

Calculate the values of m and n . (4 marks)

11 Change of subject of a formula

Equations in solving (Powers, roots and logarithms)

2012-4.

Make q the subject of the formula $a - \frac{3}{q^2} = p$. (6 marks)

2011-6.

Make y the subject of the formula $p = \sqrt{\frac{y-a}{y+1}}$. (5 marks)

2010-13.

Make a the subject of the formula: $x = \frac{\sqrt{a}}{y} + b$. (4 marks)

2009-18.

Make r the subject of the formula $P = m \left(\frac{r}{x}\right)^3$. (4 marks)

2008-4.

Make t the subject of the formula $M = K + \frac{3y^2}{t}$. (4 marks)

2007-5.

Express b in terms of a and c in the formula $c = ab - b/a$. (4 marks)

2005-9.

Make r the subject of the formula $S = \pi(2r)^2$. (4 marks)

2004-7.

Make y the subject of the formula $\frac{y+d}{c} = \frac{3d}{y-d}$. (5 marks)

2003-11.

Make m the subject of the formula $y = \frac{m}{1+m}$. (4 marks)

12 Variations

Joint variation

2012-19.

Given that $x \propto \frac{y}{z^2}$ and $x = 12$ when $y = 2$ and $Z = 1$. Find y when $x = 15$ and $Z = 2$. (6 marks)

2011-14.

p varies directly as r and inversely as the square root of q . Given that $p = 4$ when $q = 9$ and $r = 1$, calculate q when $r = 2$ and $p = 6$. (6 marks)

2010-20.

Given that $V \propto rd$ and $V = 54$ when $r=2$ and $d=3$. Find V when $r = \frac{1}{2}$ and $d = 6$. (5 marks)

2009-15.

A quantity T is proportional to m and the square of v . When $v = 3$ and $m = 5$, $T = 90$. Calculate the value of T when $m = 2$ and $v = 10$. (5 marks)

2008-21.

Given that $q \propto \sqrt{p}$ and $p = 4$ when $q = 3$, find the value of p when $q = 15$. (5 marks)

2007-12.

P varies directly as x^3 and inversely as y . When $x=2$ and $y=4$, $P=3$. Find the value of x when $P=12$ and $y=4$. (6 marks)

2006-13.

Given that $x \propto \frac{y}{z}$. When $x = 10$, $y = 2$ and $z = 4$. Find value of x when $y = 1$ and $z = 5$. (5 marks)

2005-13.

Given that P varies as a product of q and r^2 , and that $P = 50$ when $q = 1$ and $r = 5$, find P when $q = 3$ and $r = 8$. (5 marks)

2004-4.

A quantity b varies jointly with r and t , and $b = 108$ when $r = 3$ and $t = 6$. Find an equation which expresses b in terms of r and t . (4 marks)

2003-13.

Given that x varies jointly as y and inversely as the square of z , calculate the missing value in **Table 1**. (6 marks)

Table 1

x	y	z
3	1	2
1	3	

13 Polynomials

2011-10.

Given that $(x + 1)$ and $(x - 3)$ are two factors of the polynomial $ax^3 + bx - 6$, calculate the values of a and b . (7 marks)

2010-10.

Show that $k + 3$ is a factor of $k^3 + 3k^2 - 4k - 12$. (4 marks)

2009-8.

Given that $(x + 3)(x + 1)^2 \equiv Ax^3 + Bx^2 + Cx + D$, find the value of C . (4 marks)

2007-13.

When the polynomial $x^3 + 5x^2 + Kx + 3$ is divided by $(x + 2)$ it gives a remainder of 1. Find the value of K . (5 marks)

2005-20.

Given that $(4x^2 - 9)(Bx + C)$ is identical to $16x^3 + 24x^2 - 36x - 54$, calculate the values of B and C given that they are all positive. (5 marks)

2004-18.

Use the remainder theorem to prove that $(x - y)$ is a factor of a polynomial $x^2(y - 2) + y^2(2 - x)$. (5 marks)

2003-23.

Find the remainder when $2x^3 - 13x^2 - 8x + 12$ is divided by $2x - 1$. (4 marks)

14 Coordinate geometry

Gradient of a line & Equation of a line

2012-7.

Find the gradient of a straight line whose equation is $y - 1 = mx$ and passes through $(-2, -5)$. (3 marks)

2011-13.

A straight line which passes through $(3t, 7)$ and $(t, -5)$ has gradient 3. Find the equation of the line. (6 marks)

2010-9.

Figure 2 shows a straight line passing through a point $P(3,4)$.

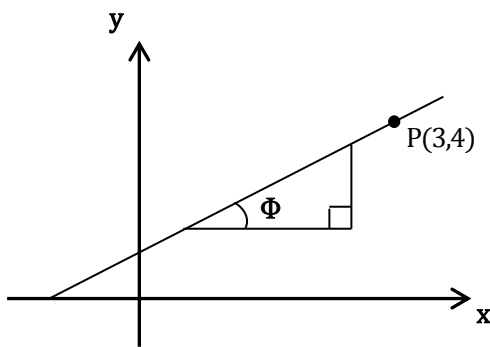


Figure 2

Given that $\tan \theta = \frac{2}{3}$, find the equation of the line in the form $y = mx + c$. (4 marks)

2009-10.

Find the gradient of a straight line whose equation is $\frac{y+2x}{4} = \frac{x}{3}$. (4 marks)

2009-13..

Given that points $P(-1,4)$, $Q(3,0)$ and $R(2, a)$ are collinear, find the value of a . (5 marks)

2007-7.

The gradient of a straight line passing through point $P(-2,5)$ is $-\frac{1}{2}$. Find the equation of the line in the form $y = mx + c$. (4 marks)

2006-14.

Two lines **G** and **H** intersect at a point **P**. **G** passes through the points $(-4,0)$ and $(0,6)$. Given that **H** has the equation: $y = 4x - 4$, find, by calculation, the coordinates of **P**. (3 marks)

2004-2.

A straight line passes through the point $(1, 6)$ and cuts the y -axis at 4, calculate its gradient. (4 marks)

2003-12.

The line joining the points $A(3, q)$, $B(5-q, 8)$ has a gradient of $\frac{1}{2}$. Calculate the value of q . (4 marks)

Parallel lines

2008-9.

Given that $y = 2x - 3$ and $y = (b - 1)x + 5$ are graphs of two parallel straight lines, calculate the value of b . (3 marks)

2005-12.

Find the equation of a straight line passing through the point $(0, 7)$ and parallel to the line $y = 2x + 5$. (4 marks)

Triangles

2006-10.

Triangle **ABC** has vertices **A** $(-1, 2)$, **B** $(3, 3)$ and **C** $(2, -1)$. Prove that angle **BAC** = angle **ACB**. (5 marks)

15 Inequalities

Simultaneous linear inequalities in one variable

2009-21.

On the same axes, sketch the graphs of the region described by the inequalities, $2 < x \leq 5$ and $0 \leq y < 4$.
Shade the unwanted region. (5 marks)

2008-20.

On the same axes, sketch the graphs of the region described by the inequalities $y < 3$ and $y \geq -x$.
(5 marks)

2004-11.

Figure 2 shows an unshaded region bounded by three inequalities.

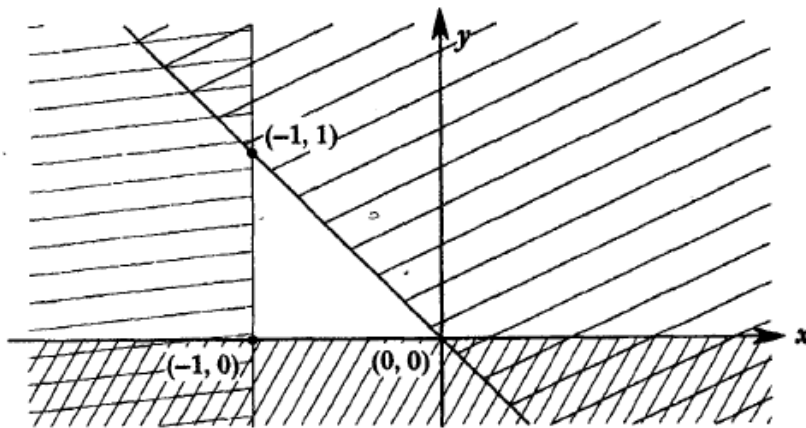


Figure 2

Write down the three inequalities. (5 marks)

Simultaneous linear inequalities in two variable

2012-18.

Figure 4 shows the unshaded region A on a Cartesian plane. Write down three inequalities that describe the unshaded region A. (6 marks)

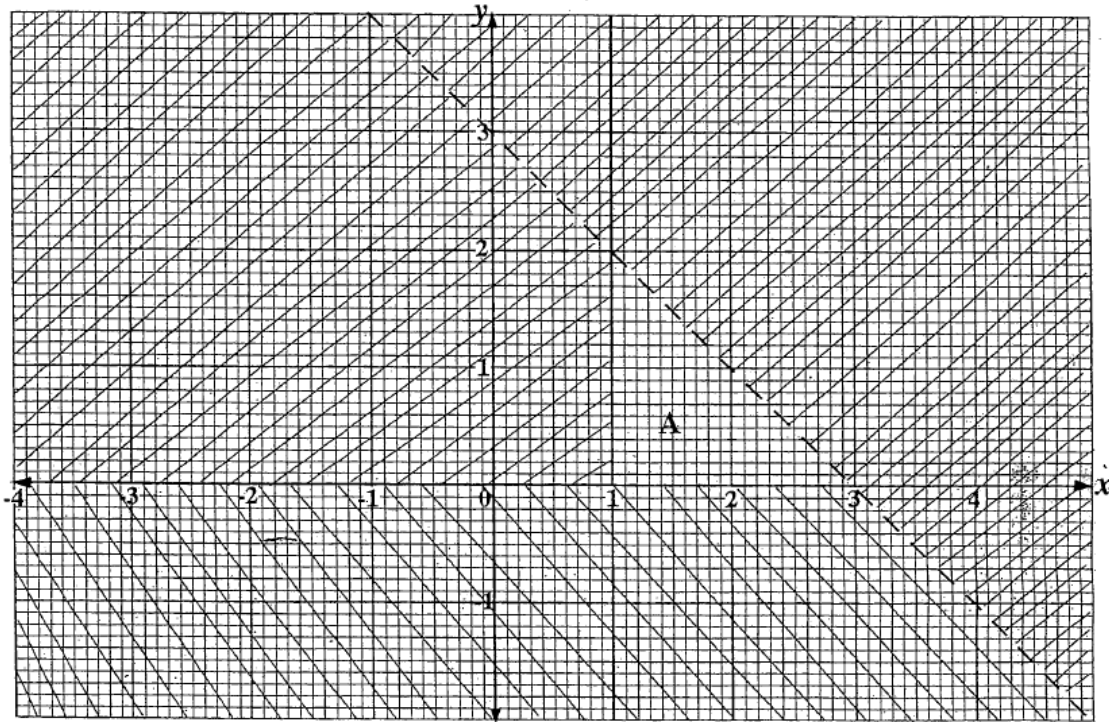


Figure 4

2011-18

Sketch the region represented by the following inequalities by shading the unwanted region: (6 marks)

$$x \geq -1$$

$$y \geq -4$$

$$y < 2x + 4$$

$$y + 2x \leq 6$$

2010-19.

On the same axes, sketch the graphs of the region described by the following inequalities:

$$x \geq 0$$

$$y = 0$$

$$y \leq 3x + 2$$

$$y + 4x < 8$$

Shade the unwanted region. (5 marks)

2007-20.

Figure 6 shows region G bounded by inequalities.

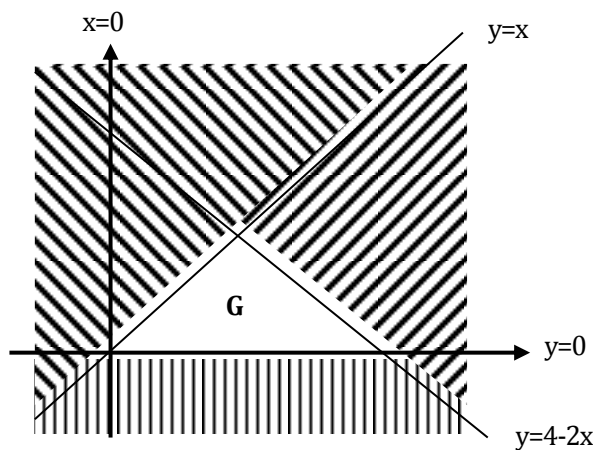


Figure 6

Write the **three** inequalities that describe the region G. (4 marks)

2005-14.

A farmer is selling at most 70 chickens out of which less than 30 are hens. Using x to represent the number of hens and y to represent the number of cocks, write down **four** inequalities involving x and y . (6 marks)

Linear programming

2007-17.

Figure 4 shows the region R bounded by three inequalities.

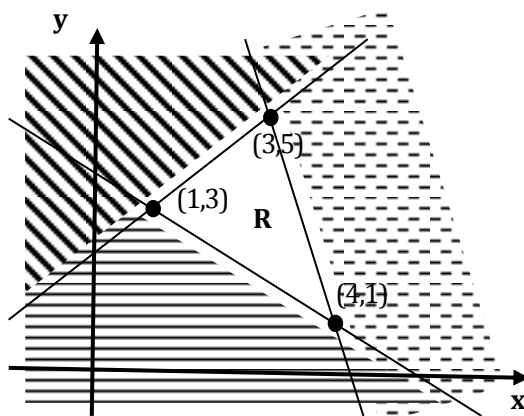


Figure 4

Calculate the maximum value of $5x - 4y + 8$ in this region. (3 marks)

2006-18.

Figure 4 shows graphs of $x + y = 8$ and $x - 2y + 4 = 0$.

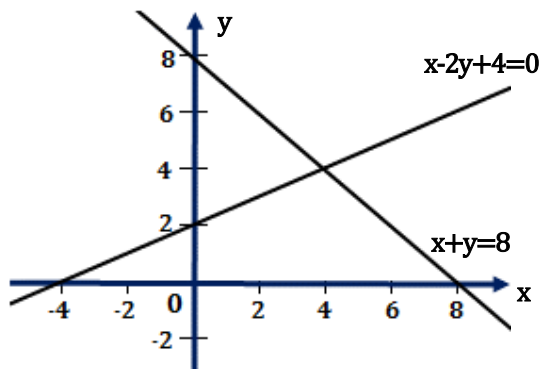


Figure 4

Copy the figure on the graph paper provided and show the region bounded by the following inequalities;

$$\begin{aligned}x &\geq 0, \\x + y &\leq 8, \\x - 2y + 4 &\geq 0,\end{aligned}$$

by shading the unwanted region. (6 marks)

2003-18.

P is a set of points (x, y) which satisfies the three inequalities:

$$\begin{aligned}x &> -1; \\y &> -2; \\x + y &< 2.\end{aligned}$$

Using a scale of 2cm to represent 1 unit on the x -axis and y -axis, draw the region **P**. (5 marks)

16 Graphs of functions

Quadratic functions

2010-16.

Figure 5 is a graph of a quadratic equation.

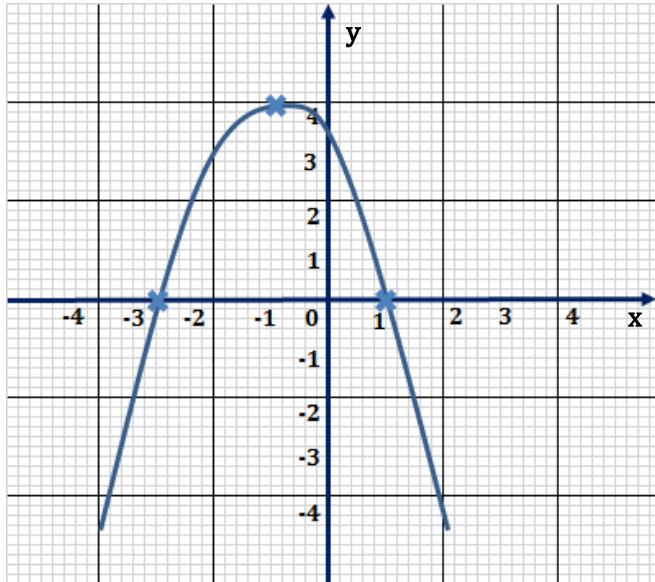


Figure 5

Formulate an equation of the graph in the form $y = ax^2 + bx + c$. (4 marks)

2009-24.

Figure 4 shows a graph of a quadratic equation $y = ax^2 + bx + c$.

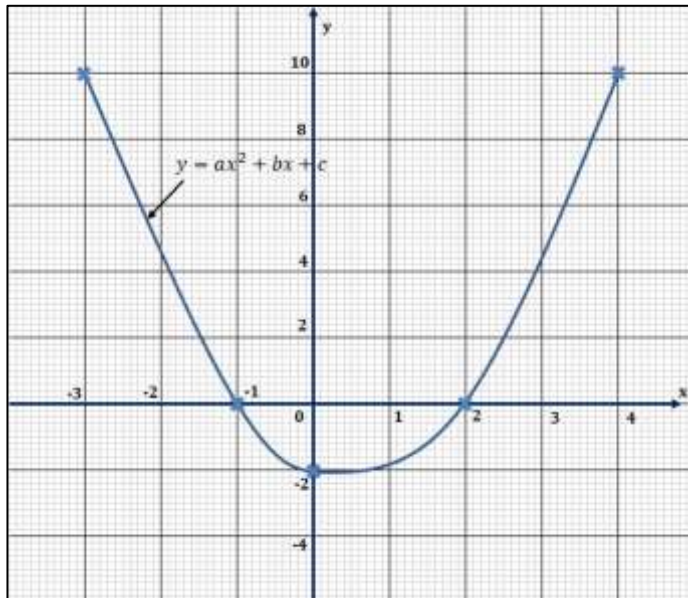


Figure 4

Find the quadratic equation for the graph in the form $y = ax^2 + bx + c$. (5 marks)

2008-13.

Table 1 shows the values of x and y of the equation $y = x^2 + x - 2$.

Table 1

x	-3	-2	-1	0	1	2
y	4		-2	-2	0	

Copy and complete the table and sketch the graph of $y = x^2 + x - 2$. (5 marks)

2005-15.

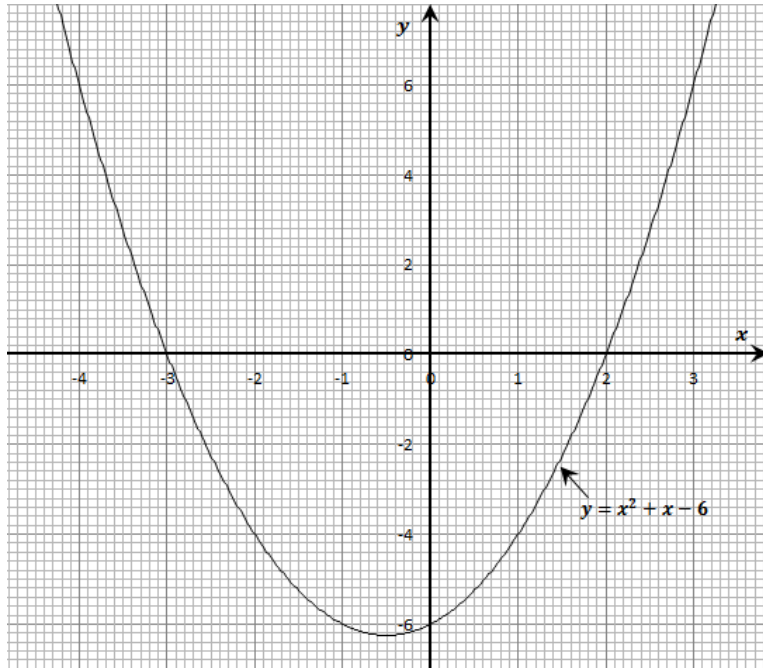


Figure 3

Use the graph to solve the equation $x^2 = 5 - x$. (4 marks)

2003-9.

Calculate the coordinates of the turning point on the curve $y = x^2 + 4x$. (4 marks)

17 Travel graph

Speed-time graph

2009-22.

Figure 3 shows a distance/time graph of an object.

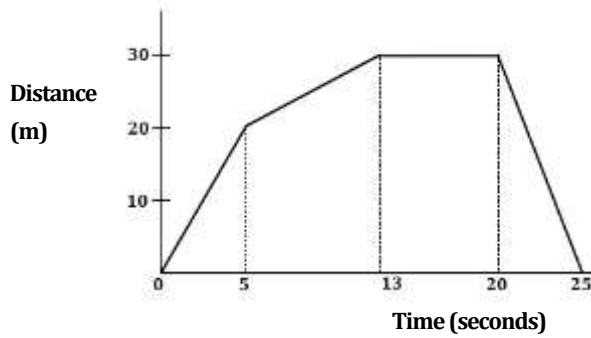


Figure 3

Calculate the speed of the object between $t = 5$ and $t = 13$. (4 marks)

Acceleration & Deceleration

2011-9.

Figure 1 is a speed-time graph of a moving object.

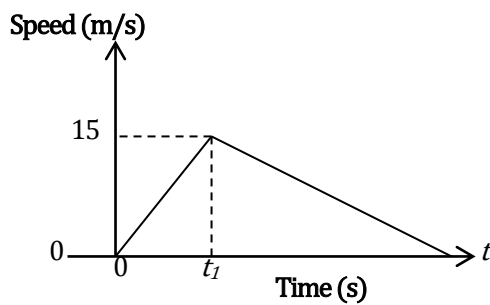


Figure 1

Given that the acceleration for the first t_1 seconds is 0.3 ms^{-2} , calculate the value of t_1 . (3 marks)

2008-16.

Figure 3 shows a speed time graph of a car.

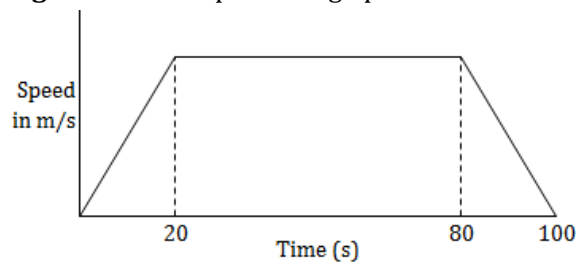


Figure 3

Given that the total distance travelled is 3200 m, calculate the deceleration during the last 20 seconds. (5 marks)

2006-12.

Table 2 shows the speed of a train recorded every 10 seconds.

Table 2

SPEED (m/s)	200	400	600	800
TIME (s)	10	20	30	40

Using a scale of 2 cm to represent 100 m/s on the vertical axis and 2 cm to represent 10 seconds on the horizontal axis, draw a speed-time graph and use it to calculate the acceleration of the train. (6 marks)

2004-12.

Figure 3 shows the speed time graph of a moving object.

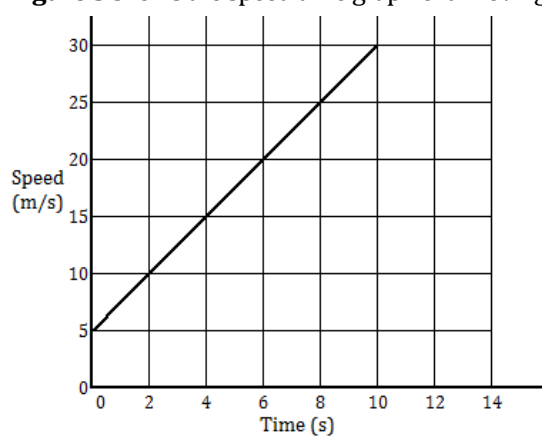


Figure 3

Use the graph to find the acceleration of the object. (3 marks)

Distance

2007-19.

Figure 5 shows the speed time graph of a moving object.

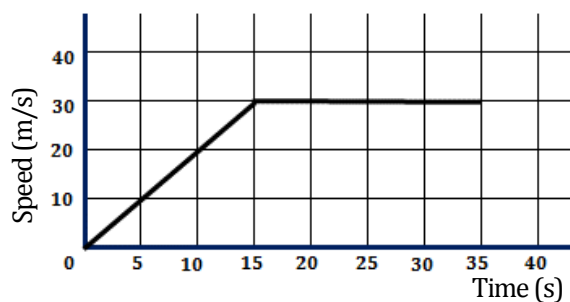


Figure 5

Use the graph to find the total distance travelled by the object in the first 35 seconds. (3 marks)

2005-19.

Figure 4 shows a speed-time graph for a car during the first 6 hours.

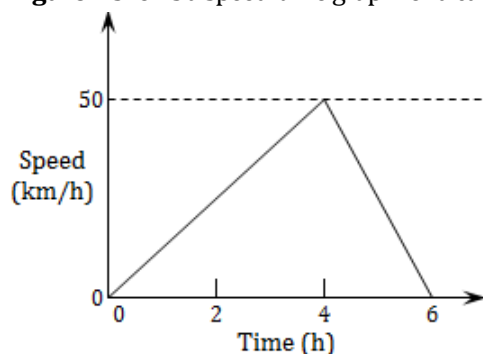


Figure 4

Calculate the total distance travelled during the 6 hours. (3 marks)

Average speed

2003-19.

Figure 1 shows a speed-time graph for a particle during the first 20 second of its motion.

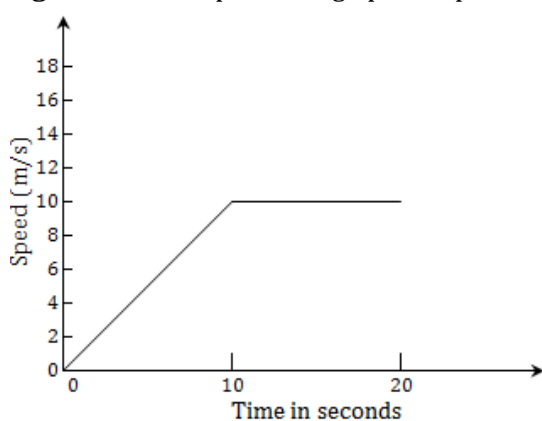


Figure 1

Calculate the particle's average speed during the 20 seconds. (5 marks)

Drawing a speed-time graph

2012-14.

A train accelerates uniformly from rest and reaches a speed of 20 m/s in 5 seconds. It then continues moving at the same speed for 3 seconds after which it accelerates constantly for 4 seconds and reaches a speed of 40 m/s. Using a scale of 2 cm to represent 10 m/s on the vertical axis and 2 cm to represent 2 seconds on the horizontal axis, draw a speed-time graph to represent the motion of the train. (5 marks)

18 The circle

Chord properties of a circle

2012-12.

Figure 2 shows a circle **AYBX** circle **O**. **XY** cuts **AB** at **C** such that angle **ACY** = 90° .

If **OC** = 8cm and **CY** = 9cm, calculate the length of **AB**. (6 marks)

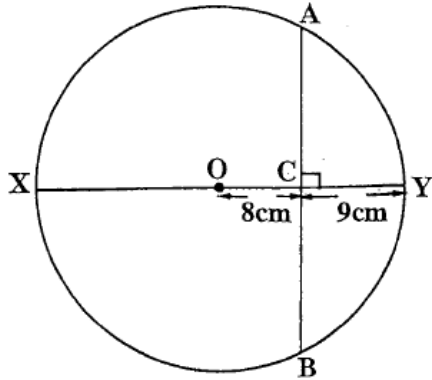


Figure 2

2010-21.

Figure 8 shows a circle **KLM** centre **O**. The diameter **KM** = 15 cm and chord **KL** = 12 cm. **E** is a point on **LM**.

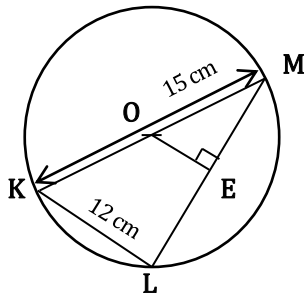


Figure 8

If **OE** is perpendicular to the chord **LM**, calculate the length of **EM**. (5 marks)

2005-17.

A chord of a circle centre **O** is 8.4cm long. If the radius of the circle is 7cm long, sketch the diagram and calculate the distance of the chord from the centre of the circle. (6 marks)

2004-13.

A chord is 6cm away from the centre of a circle. If the chord is 16cm long, calculate the radius of the circle. (5 marks)

Angle properties of a circle

2010-3.

Figure 1 shows a circle **WXYZ** centre **O**. Angle **WOY** = 112° and angle **XWY** = 36° .

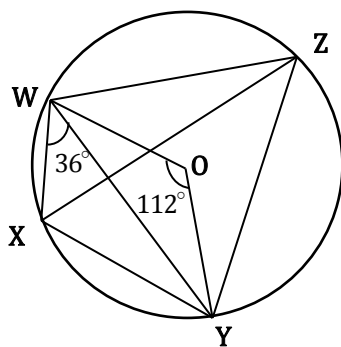


Figure 1

Calculate the size of angle **WZX**. (4 marks)

2006-19.

Figure 5 shows a circle **ABCD** in which **AB** and **DC** are produced to **G** and **F** respectively such that **GF** is parallel to **AD**.

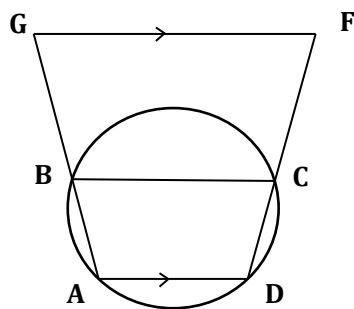


Figure 5

Prove that quadrilateral **GBCF** is cyclic. (4 marks)

2003-22.

In **Figure 3**, **ABCD** is a trapezium in which **AB** is parallel to **DC**. The diagonals **AC** and **BD** intersect at **O** such that **OD**=**OC**.

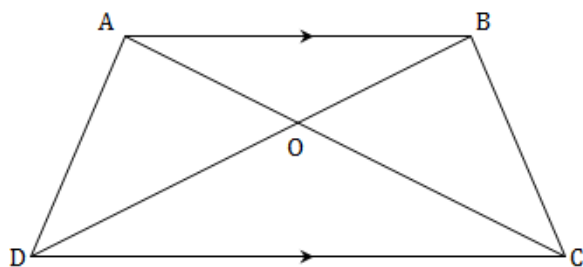


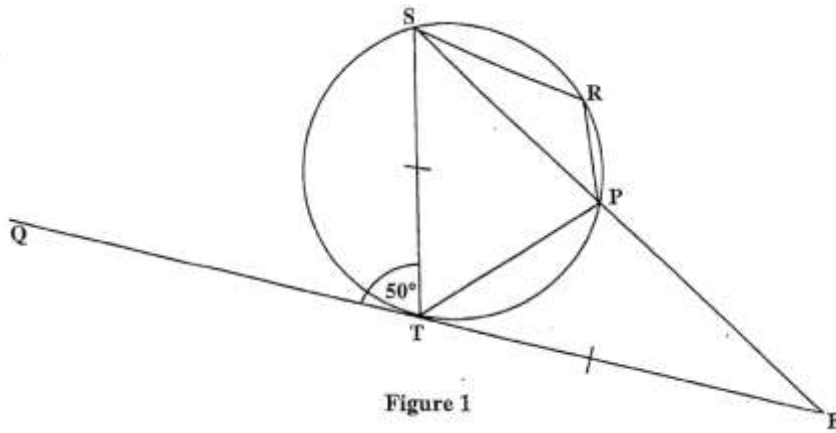
Figure 3

Prove that the points **A**, **B**, **C** and **D** are concyclic. (4 marks)

Tangent to a circle

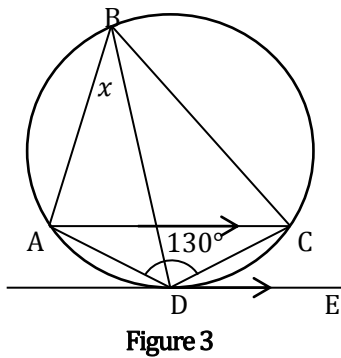
2012-9.

Figure 1 shows a tangent **QTE** to a circle **TSRP** at **T**. Angle **STQ** = 50° , and **SPE** is a straight line. If **TE** = **TS**, calculate the value of angle **SRP**. (6 marks)



2011-15.

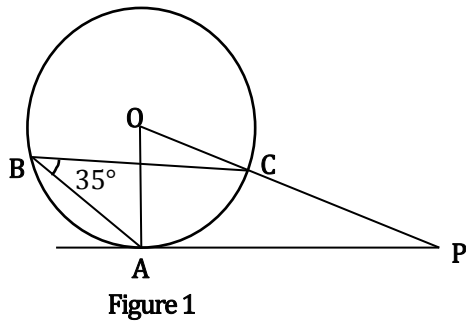
Figure 3 shows a tangent **DE** to a circle **ABCE** at **D**. **AC** is parallel to **DE** and angle **ADC** = 130° .



Calculate the value of angle **x**. (5 marks)

2009-4.

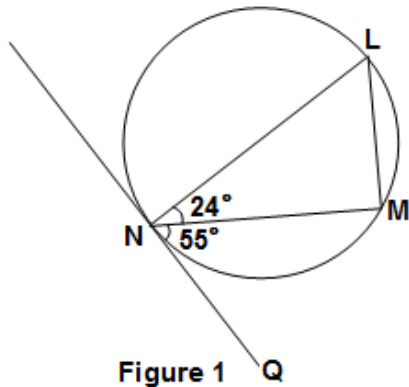
Figure 1 shows a circle **ABC** Centre **O**. **OCP** is a straight line and **AP** is a tangent to the circle at **A**.



If angle **ABC** = 35° , calculate the value of angle **APO**. (5 marks)

2008-5.

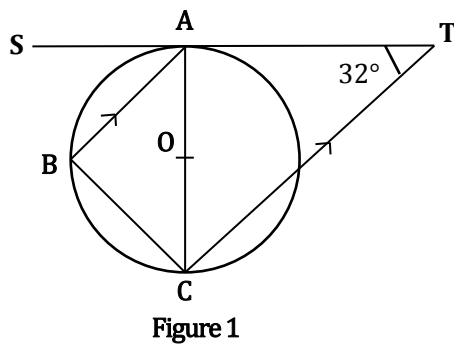
In **Figure 1** QN is a tangent to the circle LMN at N.



If angle QNM = 55° and angle LNM = 24° , calculate angle NML. (3 marks)

2007-6.

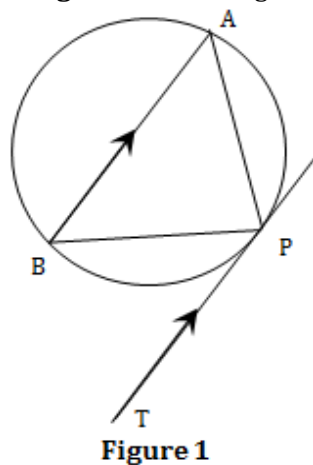
Figure 1 shows a tangent to a circle ABC with centre O. Line CT is parallel to BA and angle ATC = 32° .



Calculate angle ACB. (3 marks)

2005-3.

In **Figure 1**, TP is a tangent to the circle APB at P and AB is parallel to PT.



Prove that AP = BP. (4 marks)

2004-17.

In **Figure 4**, **TX** and **TY** are tangents to the circle **XHY** at **X** and **Y**.

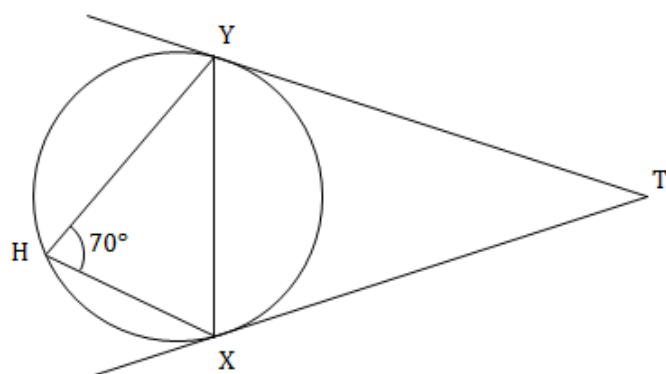


Figure 4

If angle $\text{XHY} = 70^\circ$, calculate angle XTY . (4 marks)

2003-16.

A circle centre **O** has a tangent **PA** at a point **A**. **AT** is a chord such that angle **TAP** is acute. If angle $\text{TAP} = 70^\circ$, calculate the value of angle **OTA**. (5 marks)

19 Transformations

Translation

2009-17.

Figure 2 shows line **AB**.

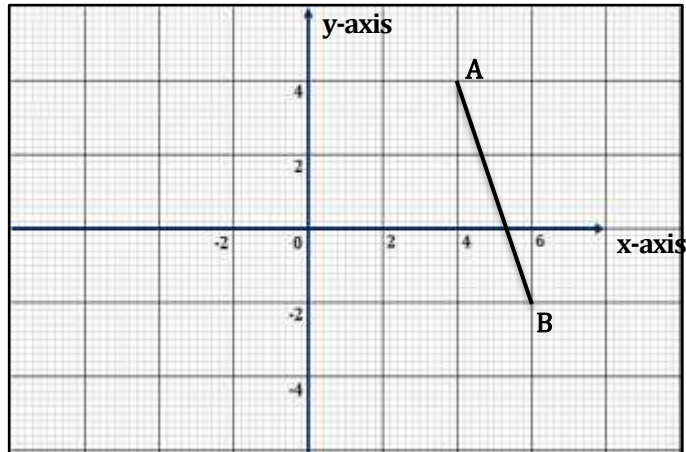


Figure 2

Calculate the coordinates of its image $A'B'$ after the translation $T = \begin{pmatrix} -6 \\ 2 \end{pmatrix}$. (5 marks)

2005-7.

A point $P(-2, 4)$ is translated to a point P' . If P' is 5 units down and 3 units to the right of point P , find the coordinates of P' . (3 marks)

Enlargement

2010-22.

On the graph paper provided, using a scale of 2 cm to represent 1 unit on both axes, draw a triangle whose vertices are $A(2,2)$, $B(3,5)$ and $C(5,3)$. On the same axes, draw the image of triangle **ABC** using $-\frac{1}{2}$ as a scale factor and $(0,0)$ as centre of enlargement. (5 marks)

2008-17.

Figure 4 shows triangle ABC and its enlargement $A'B'C'$.

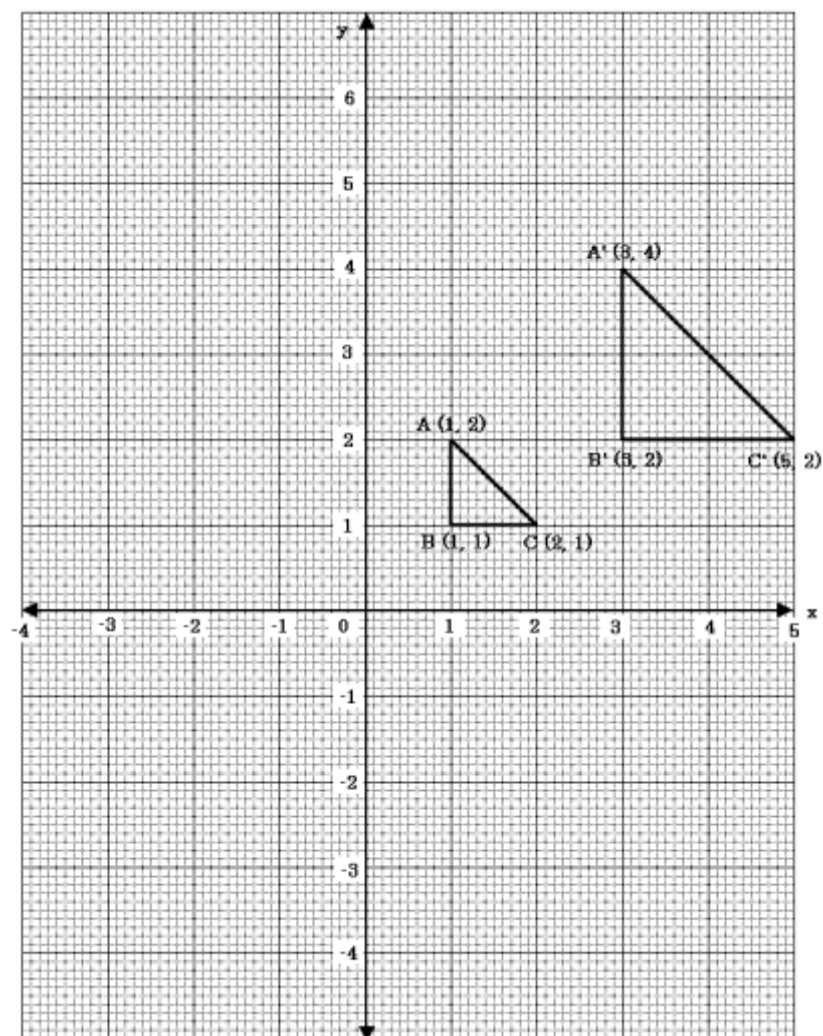


Figure 4

Copy and complete the figure to show in the same diagram the coordinates for the centre of enlargement.
(4marks)

20 Trigonometry

Sine, cosine, tangent rule

2012-15.

Figure 3 shows triangle **ABC** right angled at **B**. **BC** = 5 cm and **AC** = $\sqrt{41}$ cm. Calculate angle **ACB**, giving your answer correct to the nearest degree. (6 marks)

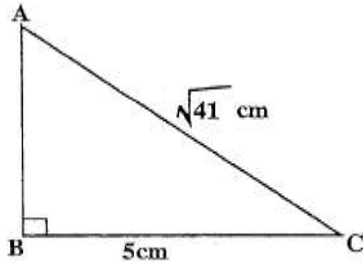


Figure 3

2011-17.

In **Figure 4**, **ABC** is a triangle such that $\angle ABC = 46^\circ$, **AB** = 7 cm and **BC** = 9 cm.

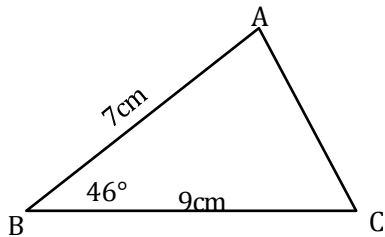


Figure 4

Calculate the length of **AC** to one decimal place. (5 marks)

2010-23.

Figure 9 shows an isosceles triangle **XYZ** with sides **XY** = **XZ** = 9 cm and **YZ** = 12 cm. **XD** is a perpendicular bisector of **YZ**.

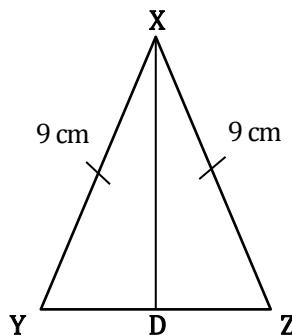


Figure 9

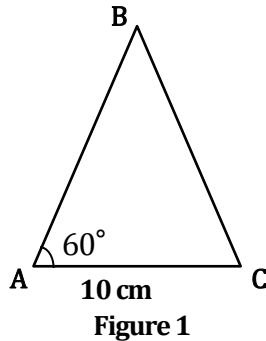
Calculate angle **YXZ**, giving your answer correct to the nearest degree. (6 marks)

2009-5.

Without using a calculator or four figure tables, evaluate $\frac{\tan 60^\circ}{\tan 30^\circ}$. (4 marks)

2006-5.

Figure 1 is a triangle **ABC** in which angle **BAC** = 60° , **AC** = 10 cm and the area of the triangle is $15\sqrt{6} \text{ cm}^2$.



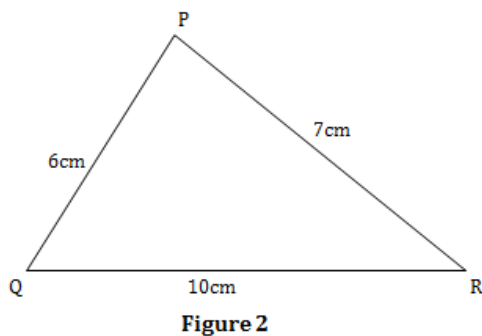
Calculate the length of **AB** leaving the answer in its simplest surd form. (5 marks)

2006-9.

Without using a calculator or four figure tables, find $\cos B$ if $\sin B = 0.8$. (6 marks)

2005-10.

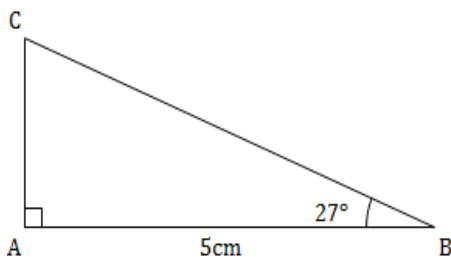
In **Figure 2**, **PQR** is a triangle such that **PQ** = 6cm, **QR** = 10cm and **PR** = 7cm.



Calculate angle **PRQ** to the nearest degree. (5 marks)

2004-6.

In **Figure 1**, **ABC** is a triangle in which angle **BAC** = 90° , angle **ABC** = 27° and **AB** = 5cm.



Calculate the length of **AC**. (4 marks)

2003-20.

A chord of a circle of radius 5cm is 8cm long. Sketch the diagram and calculate the angle subtended by the chord at the centre of the circle. (6 marks)

Angles of elevation & depression

2008-12.

An aeroplane takes off at an angle of 23° to the horizontal ground and flies for 13 km. Calculate its height above the ground, leaving your answer correct to 1 decimal place. (5 marks)

2007-21.

The angle of depression of a car from the top of a pole is 35° . If the top of the pole is 25 m from the ground, calculate the distance of the car from the pole. (6 marks)

21 Similarity

Area factor

2012-8.

The areas of two similar triangles are 75 cm^2 and 243 cm^2 . If the base of the smaller triangle is 30 cm and that of the bigger triangle is $b \text{ cm}$, calculate the value of b . (5 marks)

2011-5.

The ratio of two circles is $4:9$. Given that the radius of the bigger circle is 18 cm , find the radius of the smaller circle. (4 marks)

2010-12.

Figure 3 shows two similar triangles **TQU** and **RSU**. $TU = 3 \text{ cm}$, $UR = 6 \text{ cm}$ and **RS** is parallel to **QT**.

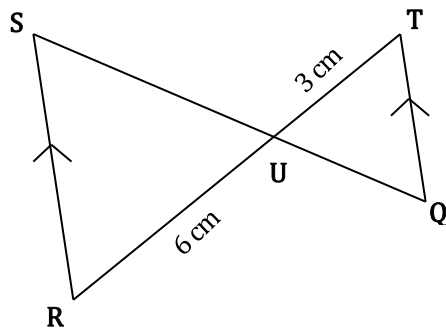


Figure 3

Calculate the ratio of the area of triangle **TQU** to the area of triangle **RSU**, leaving your answer in its simplest form. (4 marks)

2009-20.

The areas of two similar triangles **ABC** and **XYZ** are in the ratio $1:16$. If the height of the smaller triangle is 2 cm , calculate the height of the bigger triangle. (4 marks)

2007-23.

In **Figure 7**, triangle **ABC** is similar to triangle **DBA**. The area of triangle **DBA** is 24 cm^2 , $AB = 8 \text{ cm}$ and $DB = 4 \text{ cm}$.

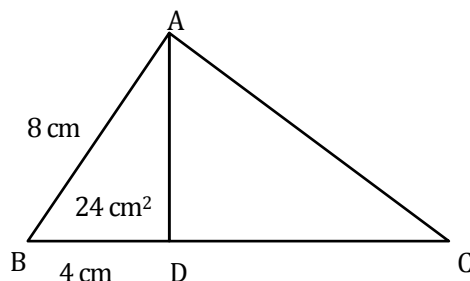


Figure 7

Calculate the area of triangle **ABC**. (4 marks)

2005-11.

A trapezium has a height of 3cm and its area is 6cm^2 . Calculate the area of a similar trapezium with a height of 12cm. (4 marks)

2004-10.

The areas of two similar triangles **ABC** and **HKL** are 100cm^2 and 256cm^2 respectively. If the length of **AB** is 5cm, calculate the length of **HK**. (5 marks)

22 Vectors

2012-6.

If $M = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and $N = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ are position vectors, find MN . (4 marks)

2011-20.

Given that $\underline{a} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$, calculate $|\underline{a} + \underline{b}|$. (6 marks)

2010-14.

Figure 4 shows a quadrilateral **KLMN** in which $\overrightarrow{NK} = 3\underline{d}$, $\overrightarrow{NM} = \underline{e}$ and $\overrightarrow{ML} = \underline{d}$.

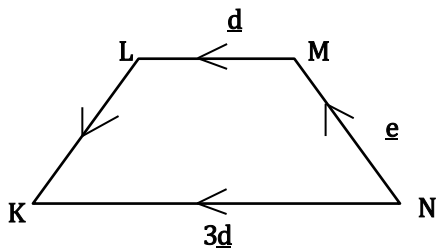


Figure 4

Express $\overrightarrow{LK}$ in terms of $\underline{d}$ and $\underline{e}$. (4 marks)

2008-23.

In **Figure 5**, $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and M is the mid-point of $\overline{AB}$.

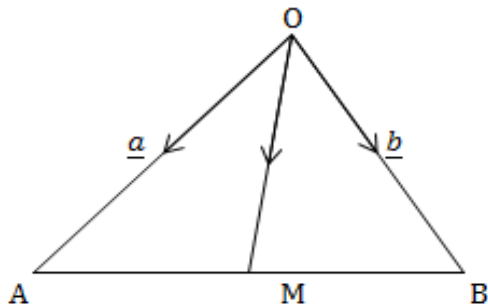


Figure 5

Find $\overrightarrow{OM}$ in terms of $\underline{a}$ and $\underline{b}$. (5 marks)

2007-2.

Given that $\underline{a} = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$, find $\frac{1}{2}(\underline{b} - \underline{a})$. (4 marks)

2007-16.

Figure 3 shows the image P' of P after translation.

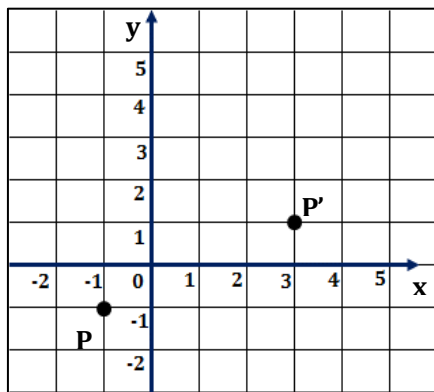


Figure 3

Find the translation vector that maps P into P' . (3 marks)

2006-24.

Figure 6 shows a parallelogram $ABCD$ in which $\overrightarrow{AB} = \underline{x}$ and $\overrightarrow{BC} = \underline{y}$.

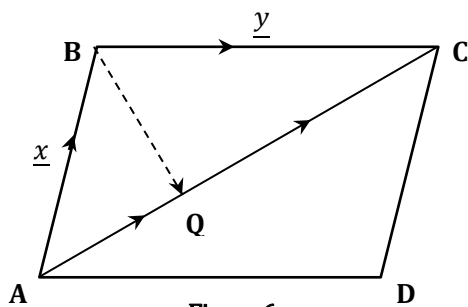


Figure 6

If $\overrightarrow{AQ} = \frac{1}{4}\overrightarrow{AC}$, find $\overrightarrow{BQ}$ in terms of $\underline{x}$ and $\underline{y}$. (5 marks)

2005-22.

Given that $\overrightarrow{AB} = \begin{pmatrix} -9 \\ 4 \end{pmatrix}$, calculate the length of $\overrightarrow{AB}$, leaving your answer correct to 3 significant figures.

(4 marks)

2004-5.

Given that $C' = (5, -5)$ is the image of $C(1, 1)$ after translation, find the translation vector that maps C into C' . (4 marks)

2004-15.

Given that $\overrightarrow{AB} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 9 \\ 15 \end{pmatrix}$, show that points A , B and C are collinear. (3 marks)

2003-7.

Calculate vector $\overrightarrow{AB}$ if vectors $A = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$ and $B = \begin{pmatrix} -4 \\ 4 \end{pmatrix}$. (3 marks)

23 Three dimensional figures

Surface area and volume

2012-20.

Figure 5 shows a solid cone of base radius $OC = 7\text{cm}$ and the slant height $= 21\text{cm}$. Calculate the total surface area of the cone. (Curved surface Area of cone $= \pi rl$, take $\pi = \frac{22}{7}$) (6 marks)

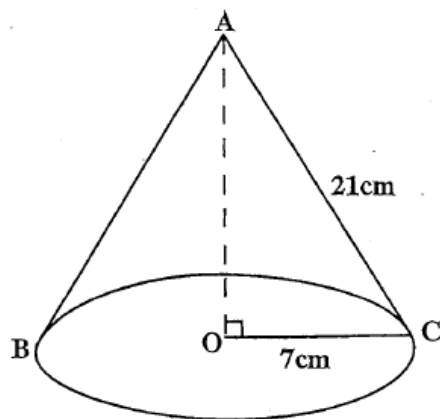


Figure 5

2011-19.

A pond 12m in diameter, has a shape of a hemisphere and is full of water. The pond is emptied and all the water poured into a cylindrical tank of radius 5m. Assuming there is no loss of water, calculate the height of water in the tank. (volume of sphere $= \frac{4}{3}\pi r^3$) (6 marks)

2009-3.

The volume of a pyramid is 60cm^3 and its base area is 20cm^2 . Calculate the height of the pyramid. (Volume of pyramid $= \frac{1}{3} \text{base area} \times \text{height}$). (3 marks)

2008-15.

A cylindrical metal bar whose volume is 594cm^3 is melted and cast into a sphere. Calculate the radius of the sphere, leaving your answer correct to 2 decimal places. (Volume of a sphere $= \frac{4}{3}\pi r^3$, Take $\pi = 3.142$). (5 marks)

2007-18.

Calculate the total surface area of a solid hemisphere of radius 21cm.

(Area of a sphere $= 4\pi r^2$; Take $\pi = \frac{22}{7}$). (5 marks)

2006-3.

A cuboid is 76 cm long, 50 cm wide and 40 cm high. Calculate the volume of the cuboid. (3 marks)

2004-21.

Figure 6 shows a triangular prism, in which $BC = 3\text{cm}$, $EF = 4\text{cm}$, $BF = 6\text{cm}$ and angle $ABC = 90^\circ$.

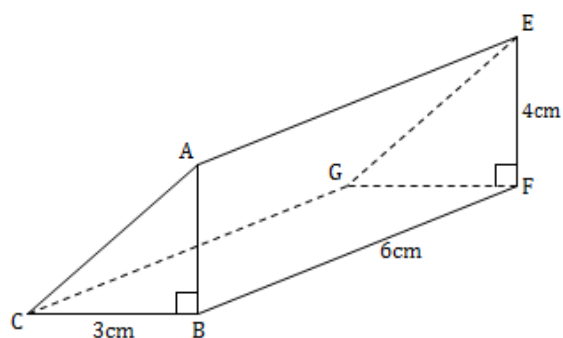


Figure 6

Calculate the volume of the prism. (3 marks)

Sides in 3-D figures

2005-23.

Figure 6 shows a right-cone whose vertical angle $BAC = 116^\circ$, the diameter of its base $BC = 176\text{mm}$ and AX is the height.

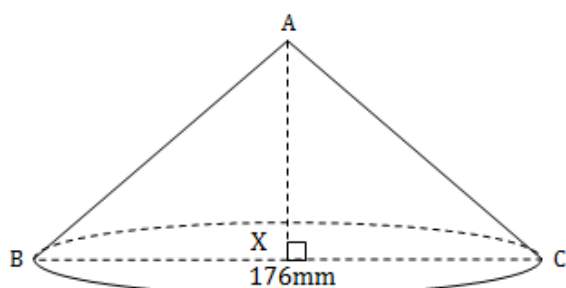


Figure 6

Calculate the length of AX . (5 marks)

Angles

2010-18.

Figure 7 shows a rectangular based right pyramid $FGHIJ$, $GH = 10\text{ cm}$ and $HI = 7\text{ cm}$.

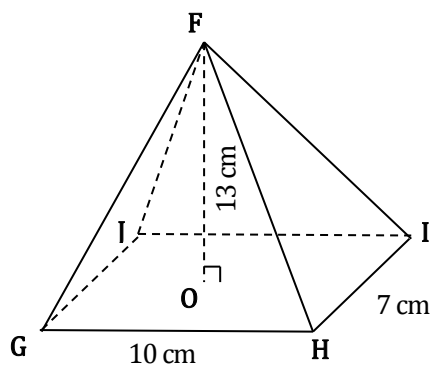


Figure 7

If the height of the pyramid is 13 cm , calculate the angle between FHI and the base. (5 marks)

24 Constructing geometry

Tangent

2010-24.

Using a ruler and a pair of compass only, construct in the same diagram, (6 marks)

- (i) A circle centre **O** of radius 4 cm.
- (ii) A tangent **TP** to the circle at any point **P** such that angle **POT** = 60° .
- (iii) Measure and state the length of **PT**.

2008-22.

Using a ruler and a pair of compasses only, construct in the same diagram: a circle centre **O** of radius 3 cm, a tangent to the circle at **P** from a point **T** which is 8 cm from the centre of the circle. (4 marks)

2003-24.

Draw a circle centre **O** with radius 3cm. Construct another circle radius 4cm passing through point **O**. Label its centre **C**. Label one of the intersection points of the two circles **A**. Using a ruler only, construct a tangent to the circle centre **O** at point **A**. Measure and state angle **AOC**. (5 marks)

Bisect perpendicular

2006-16.

Using a ruler and pair of compasses only, draw a line **AB** = 10 cm and construct a circle with the line **AB** as a diameter. Mark a point **C** on the circle such that **AC** = 6 cm. Join **AC** and **BC**. Construct a tangent **CP** such that angle **BCP** is acute. (5 marks)